Prediction Markets: the "Oracle Problem" is not the problem.

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For prediction markets, The Oracle Problem is *not* the actual problem, perhaps unintuitively. Stop blaming UMA.

I'll explain. The so-called "Oracle Problem" is one of the oldest ones in crypto; like everything, it goes back to the early days of bitcointalk forum.

And it is always at the center of discussions around prediction markets *(can I offer you some Truthcoin? a 2014 example)*: oracles are expected to report the truth onchain--according to real world outcomes--and have onchain financial programs rely on their report.

The Oracle Problem has never been easy, but for prediction markets in particular, it appears to be even harder for a few reasons. The most obvious one; a malicious oracle report is catastrophic! the wrong contract goes to 0, causing a complete loss of funds for a big set of users. For the less obvious reason, well, keep reading for another minute.

Polymarket had many controversial oracle rulings, and some extremely contentious to the point of users feeling like they were straight-up cheated:

1. Will Zelensky wear a suit before July 1, 2025
2. Israeli military action in Syria
3. Who wins the 2024 Venezuela presidential election
4. LayerZero's airdrop
5. Barron's involvement in Shkreli's Trump memecoin
6. Ethereum ETF Approval
7. Is Sam Bankman-Fried jailed
8. Will Trump tweet

9. Israeli invasion of Lebanon
10. Gold missing from Fort Knox
11. Will the Titan submersible be found
12. Will the US and Ukraine sign a mineral mining deal before EOM
... I could go on.

Each of these markets had its own unique story and its own set of users needlessly burnt. In a few of them, many millions of millions USD were on the line. Often, the majority of the bets piled in **after** the event in question happened, just to speculate on which way the UMA oracle would decide to rule the event.

Pretty f*ed right? How do we fix this?

Naturally, as in crypto we are, our minds instantly lean to: *"OK, to solve this we have to make oracles trustless and decentralized too, right!?"*

Well no, actually. In the context of prediction markets, that doesn't do much. May sound weird at first, but bear with me.

We have to introduce important distinctions between the normal oracle feeds we're used to, and *event oracles*:

A) Price feeds *(normal):* liquid price is a known definition, and the truth is widely agreed upon. Contention about the price of a liquid asset is rare - as its definition is widely understood for hundreds of years; the price is typically the **last traded price** or the **bid-ask midpoint** in an active market.

Even *in extreme cases* where a few exchange books become empty or distorted, the real market price for these liquid assets is clear to participants (with minor noise variation). And furthermore, even in high sigma scenarios where all books are so empty that market price isn't clear-- it's a temporary situation, and oracle price feeds can usually be paused and resumed shortly after, without critical repercussions to defi protocols (as happened before). **Tldr** price feed oracles worry about liveness, and about weeding out malicious reports. But not about defining what *'price'* means.

B) Event feeds: the event oracle doesn't need to worry about liveness. It does need to worry about malicious reporting. But most importantly, it has to

concern itself with **definitions**. You see, for price feed oracles, the definition of a "market price" is implicitly known, understood, and permanent. However- the definition of events in prediction markets is **man-made**, and wildly **differs between each event**. For event oracles; the question of the ***truth itself*** is often contested- on a most fundamental level.

In reporting event outcomes, the oracle problem becomes trivial compared to the **definition problem**. Prediction markets can have the most honest, trustless, decentralized and neutral oracle conceivable-- but if there is a fundamental logic dispute on which outcome SHOULD be reported, then a decentralized oracle is like serving cake icing without the cake – pointless.

In other words; since crypto's inception, oracles have been designed to report truth correctly, but never to decide what the truth actually *is.*
When the assumption of clear definition breaks down, an oracle simply becomes useless.

Let me relax, and let's take a step back for a second.

How did we get here?

Understandably, oracles have always been designed with the assumption of clear definitions. There was demand for price feeds, TVL, interest rates, sequencer uptimes; and other metrics **all of which are empirical, reproducible data feeds.**

The principle described in a 2014 blog post called SchellingCoin: A Minimal-Trust Universal Data Feed, Vitalik laid out what is essentially still the foundation of all oracle protocols 11 years later: if everyone knows what the schelling point value is, such that actors are confident what other actors are likely to report, the protocol can reward actors who report close to everybody else, and slash outlier reporters, as they can be assumed to be malicious or lagging.

Alas, **the whole design breaks down if there is no obvious schelling point.**

All that to say; for prediction markets. The Oracle Problem is not really the

oracle problem, because crypto oracles were never designed for such things as world events.

Now, if trustlessness and decentralization don't matter much for prediction market oracles because the most common contention is definition disputes... What *does* matter? How *can* we improve it?

Recognize these two practical guiding principles:

1) The oracle actor(s) need to have a direct stake in the long term success of the platform. This is a basic principle well known in consensus mechanisms that for some reason has gone ignored (all PoS, PoW designs acknowledge this need). A proprietary Polymarket centralized oracle is preferable to an external “deCenTRraLizEd” UMA oracle composed of mercenary capital.

2) Definitions definitions definitions! The oracle is secondary to providing an airtight definition of events. In almost all of Polymarket’s (many) controversies, a continuous carelessness towards definitions was at fault.

Moreover, acknowledge that if the real world is imperfect; even best-efforts in writing an initial definition might miss a possible edge case. In that case, immediate clarifications should be provided to stop the issue from snowballing further.

GIF

And most importantly; snub the people trying to sell you on complex, denetralized oracle mechanisms . I guarantee you, they didn't solve the *real* issue plaguing prediction markets. You are now armed with the ammunition to tell them to f*k off.

Lastly, if you’re interested in markets and frontier financial protocols, we *(stealth)* are looking for a fullstack dev that can bring React, Tanstack, and Websockets to their knees and make users gasp *“wow numbers move fast on*

my screen”.

p.s. we’re not building a prediction market, let those 200 products compete between themselves while we build the next big thing.

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